# 4. Comparison with Baselines

M3 Research Project · Detection-Constrained Power Allocation for Cognitive ISAC Relays under Imperfect CSI (NCC 2027 submission)

Source: submitted NCC 2027 paper, Sec. V-A, Table II, Figs. 2-4 and Sec. V-E.

## Compared schemes (Table II)

| Scheme | Radar rule | CSI errors | Default |
|---|---|---|---|
| **Proposed** | $P_D \geq \beta$ | averaged | $\beta = 0.9$ |
| Hu non-robust [3] | SINR $\geq \eta_{min}$ on estimated channels | ignored | 10 dB |
| Hu robust [3] | SINR $\geq \eta_{min}$ for every error in a bounded set covering 95 % of the Gaussian error on each link | worst case | 10 dB |
| Adapted worst-case | our alternating design with a worst-case SINR floor $\eta$ | worst case | 9.80 dB (the floor equivalent to $\beta = 0.9$, eq. 7) |

The baselines cover the two usual ways of protecting sensing:

- an SINR floor on the estimated channels [4], [5];
- a worst-case guarantee over a bounded error set [3], [7].

## Fairness of the comparison

- All schemes use the settings of Table I and the **same channel realisations**. They differ only in how the radar is protected.
- **Tuned floors.** Both Hu schemes are also run with a tuned floor: the smallest floor that protects at least 95 % of 20 separate development channels. This keeps the benchmarks from being limited by a poorly chosen setting.
- **One evaluator.** A single evaluator scores every design, using the received-signal model and a threshold calibrated to $P_{FA} = 10^{-3}$.
- **Own design parameter.** In Fig. 3, each scheme is swept over its own design parameter and compared at a common protection requirement.

## Summary of the comparison (ε = 0.01 unless stated)

| Comparison | Proposed | Hu robust [3] | Hu non-robust [3] | Adapted worst-case |
|---|---|---|---|---|
| Protected SAR, default settings (Fig. 2, 60 ch.) | **8.37** | 4.76 (tuned 5.54) | protects few channels | 5.86 |
| Protected SAR, ε = 0.03 (Fig. 2) | **6.57** | 3.04 | tuned: 0 | 4.43 |
| Avg. SAR at 100 % protection (Fig. 3, 40 ch.) | ≈ **8.4** (β = 0.9) | ≈ 5.8 (4 dB) | ≈ 5.3 (19 dB) | ≈ 7.3 (4 dB) |
| Worst-target P_D behaviour (Fig. 4) | many close to 0.9 | ≈ 1 | untuned: many below 0.9; tuned: ≈ 1 | ≈ 1 |
| Per-error-draw protection (Sec. V-F) | 71 % | 100 % | — | 100 % |
| Worst-case SAR over error sets (Sec. V-F) | 2.30 | **2.86** | — | — |

**Gains.**

- Fig. 2: **+76 %** over Hu robust.
- Fig. 3: **+15 %** over the best benchmark at full protection.
- Component study (Sec. V-E): **+39.6 %** over the worst-case SINR rule, with the allocation procedure held fixed.

**Where the baselines do as well or better.**

- For $\varepsilon <= 1.5 \cdot 10^{-3}$, the tuned Hu non-robust floor comes within 3 % of the proposed design.
- On the worst-case criterion, Hu robust is higher: 2.86 against 2.30 bit/s/Hz.